Image Reconstruction using Self-Prior Information for Sparse-View Computed Tomography

Mona Selim, Essam A. Rashed, Mohammed A. Atiea and Hiroyuki Kudo

I. Abstract

The contradiction between the great benefits of computed tomography (CT) in diagnosis and the risk of redundant CT scan on the patient health, make the researchers compete developing image reconstruction methods for low-dose CT. Sparse-view CT is a common technique in radiation dose minimization. Due to the streak artifacts that result while using the analytical reconstruction method with sparse-view CT, several iterative reconstruction methods have presented to produce high image quality. In this work, we introduce extracting the prior information incorporated in the reconstruction method during the process of reconstruction itself, in contrast to the other related methods that prepare the prior information in advance. The proposed technique is divided into two main steps. The first step is the construction of self-prior information. The second step is incorporating this produced information into the reconstruction process. The performance of the proposed method is evaluated using simulation and synthetic real data. Results show that the proposed technique produce high image quality.

Keywords—computed tomography, image reconstruction, sparse-view, self-prior.

II. Introduction

Computed tomography (CT) is used extensively in medical diagnosis as well as image-guided radiation therapy. However, it is known that the frequent scan with CT might lead to malignancy and cancer, which cause a risk on the patient health [1]. Subsequently, there is an increased interest in decreasing this risk on the patient by minimizing the x-ray CT radiation dose. Many techniques have been used to minimize the radiation dose by adjusting the x-ray tube power to produce lower x-ray dose than normal dose. Furthermore, CT scan with small number of views, which also called sparse-view, has the ability to diminish the radiation dose. Unfortunately, implementing the conventional analytical reconstruction method, such as filtered back-projection (FBP), with dose minimization techniques produces images with noise and streak artifacts. Consequently, CT will lose its main task in diagnosis. Due to the benefits of CT in diagnosis and the risk of frequent CT scan on the patient health, researchers compete developing image reconstruction methods that are able to produce high quality images with low-dose techniques. In this work, we consider using sparse-view as a way to diminish the x-ray radiation dose. Based on the theory of compressed sensing [2], many iterative reconstruction methods have been developed for sparse-view CT. Among them, the methods that merge prior knowledge about the reconstructed image into the image reconstruction process. Many examples of the prior models were introduced to deal with sparse-view CT problems such as intensity prior [3], total variation (TV) prior [4] and Gamma regularization [5]. The complete reference image considers one form of this prior information, and it may be obtained from either a scan of the current patient or other patients [6], [7]. Also, the reference image can be obtained from different phase in the case of multiphase abdominal CT images [8]. In all of these cases, the prior information or the reference image must be prepared earlier to the reconstruction process. Because the prior information come from different scan, the patient size, position or orientation is not necessarily to be the same. Hence, a registration process between the reconstructed image and the reference image must be performed before the reconstruction process. The quality of such methods depends on the accuracy of the registration process. If the images are well registered, these methods will produce high image quality and vise versa. Therefore, our main goal in this work is to extract the prior information or the reference image during the reconstruction

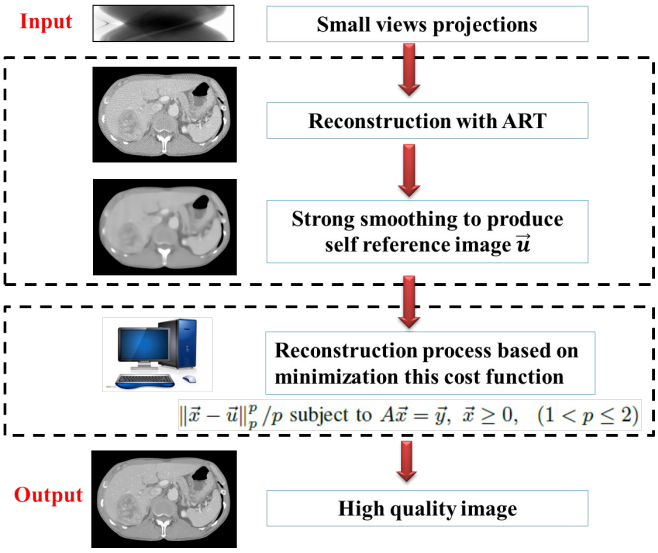

Fig. 1. Overview of the proposed technique.

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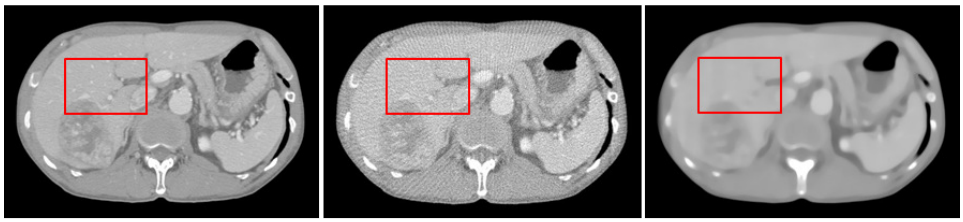

(a) (b) (c)

Fig. 2. Example of strong smoothing with TV. (a) Central slice of abdominal CT image, (b) Reconstructed image using ART algorithm from sparse-view and (c) the resultant image from strong smoothing with TV minimization. It contains the same structures of (a) but the small size details such as those in the red rectangle are difficult to observe.

process itself to overcome the limitation of conventional methods by canceling the step of registration process. In other words, the reference image is automatically generated during the reconstruction. Furthermore, extracting self-prior information presents another benefit that there is no longer need to prepare prior information before the reconstruction process. The proposed technique is divided into two main steps. The first step is the construction of self-prior information. The second step is incorporating this produced information into the reconstruction method by adding ℓ_p norm distance between the self-produced information and the reconstructed image to the data fidelity term in the cost function. Overview of the proposed technique is shown in Fig. 1.

III. Proposed Method

A. Construction of Self-Prior Information

We propose to produce the self-prior information as follow. An initial image is reconstructed using conventional reconstruction method such as algebraic reconstruction technique (ART) [9]. We propose to use ART algorithm with small number of iterations to reduce the computational cost. This reconstructed image is expected to contain streak artifacts. Smoothing process can help in removing these streak artifacts. Normal smoothing can not remove all the artifacts, while strong smoothing will remove all the streak artifacts as well as removing small details from the image. In the proposed technique, we propose to use strong smoothing using TV minimization [10] where it is characterized by edge preserving. TV is defined in Eq. (1). Strong smoothing for the reconstructed image using ART will result in image that contains the basic structure as the original image but without any details as shown in Fig. 2b. This image that contains the basic structure is enough to be used as prior information in the reconstruction process to enhance the image quality. We call this smoothed image as reference image $\vec{u} = (u_1, \ldots, u_n)^\top$ and it will be used in the reconstruction method. As a brief description of how to produce the self-prior information, initial reconstruction with ART algorithm with small number of iterations for example. Then, strong smoothing for the reconstructed image using TV minimization.

$$TV(\vec{x}) = \sum_{s,t} \sqrt{(x_{s+1,t} - x_{s,t})^2 + (x_{s,t+1} - x_{s,t})^2 + e}. \quad (1)$$

where e is a small value.

B. Image Reconstruction Algorithm

In x-ray CT, the image reconstruction technique can be presented as solving system of linear equations as follows

$$A\vec{x} = \vec{y}, \quad (2)$$

where $\vec{x} = (x_1, \ldots, x_n)^\top$ is a vector denote the object image, $\vec{y} = (y_1, \ldots, y_m)^\top$ is a vector represent the projection data and $A = \{a_{ij}\}$ define the $m \times n$ system matrix.
We minimize the following cost function for the image reconstruction

$$\|\vec{x} - \vec{u}\|_p^p / p \text{ subject to } A\vec{x} = \vec{y},\ \vec{x} \geq 0, \quad (1 < p \leq 2) \quad (3)$$

This cost function is composed of two parts. The first part is the data fidelity term $A\vec{x} = \vec{y}$ and its role is to ensure the consistency between the reconstructed image $\vec{x}$ and the projection data $\vec{y}$. The second part is the ℓ_p norm distance between $\vec{x}$ and $\vec{u}$ and it is used to enforce image sparsity and diminish the streak artifacts. The cost function is minimized using the Lagrangian duality similar to the study in [11]. The Lagrangian dual problem correspond to Eq. (3) is defined as

$$\text{maximize } D(\vec{\lambda}) = \vec{y}^\top \vec{\lambda} - \sum_{j=1}^{n} g_{u_j}(b_j), \quad (4)$$

$$g_a(t) = \begin{cases} -\frac{1}{p}a^p & (t < -a^{p-1}) \\ \frac{1}{q}|t|^q + at & (t \geq -a^{p-1}) \end{cases}, \quad q = \frac{p}{p-1}.$$

where $\vec{\lambda}$ is Lagrangian multipliers with optimal value $\vec{\lambda}^*$. The Lagrangian dual problem in Eq. (4) is solved by the coordinate ascent method to obtain the optimal value $\vec{\lambda}^*$, then the minimum value $\vec{x}^*$ for the problem in Eq. (3) is computed with the following primal-dual relation:

$$x_j^* = g'_{u_j}(A^\top \vec{\lambda}^*|_j),$$
$$g'_a(t) = \begin{cases} 0 & (t < -a^{p-1}) \\ \text{sgn}(t)|t|^{q-1} + a & (t \geq -a^{p-1}) \end{cases}.$$

The iterative reconstruction algorithm is as follow
Step 1: Set $\vec{w}^{(0)} = (0, \ldots, 0)^\top$ and $k \leftarrow 0$
Step 2: For $k = 1, 2, \ldots, K-1$

$$\vec{w}^{(k,1)} = \vec{w}^{(k)}$$
$$\vec{w}^{(k,i+1)} = \vec{w}^{(k,i)} + \beta \frac{y_i - \sum_{j=1}^{n} a_{ij} g'_{u_j}(w_j^{(k,i)})}{\max(\Gamma, \sum_{j'=1}^{n} a_{ij'}^2 h(u_{j'}, w_{j'}^{(k,i)}))} \vec{a_i}$$
$$\vec{w}^{(k+1)} = \vec{w}^{(k,m+1)}$$

Step 3: Compute $\vec{x}^{(k+1)}$ as:

$$\vec{x}^{(k+1)} = (g'_{u_1}(w_1^{(k+1)}), \ldots, g'_{u_n}(w_n^{(k+1)}))^\top$$

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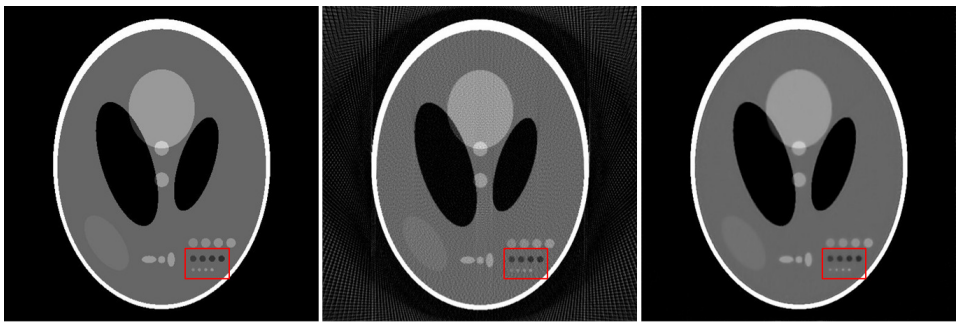

(a) (b) (c)

Fig. 3. Reconstruction results of the simulation data. (a) The modified phantom, (b) result of reconstruction using FBP algorithm, and (c) result of reconstruction using the proposed technique. The red rectangles in the images indicate that even the small ellipses are well reconstructed. The gray scale is displayed from 0 and 0.5 cm^{-1}.

where β is a hyperparameter, Γ is a positive constant, K is the iteration number. Also, $h(.)$ is an approximation of the second derivative of $g_a(t)$ and is computed as follows:

$$h(t) = \begin{cases} 0 & (t < -a^{p-1}) \\ \frac{1}{2}(q-2)a^{2-p} & (t = -a^{p-1}) \\ (q-2)|t|^{q-2} & (t > -a^{p-1}) \end{cases} .$$

IV. Experimental Results

The proposed technique is evaluated through simulation and synthetic real data.

A. Simulation data

We execute the experimental simulation on the Shepp-Logan phantom. We add small ellipses of different sizes to the phantom to represent the small anatomical details. This modified phantom was used in the simulation study for the goal of evaluation. The experiments were implemented using C++ platform on a PC (Intel i5 CPU and 8 GB RAM). The phantom resolution is 512×512 pixels and the projection data were produced using parallel beam projections. We used 128 views for the reconstruction process and implemented the proposed technique using two steps. The first step is reconstruction using ART algorithm with 10 iterations. Then, strong smoothing using TV minimization is executed. The reconstruction in the second step is implemented with these parameters: p is 1.1, β is 1.3, Γ is 256, and K is 30. Furthermore, we reconstructed the same slice using FBP algorithm to compare the proposed method with the FBP for the evaluation purpose. We used root mean square error (RMSE) and structural similarity (SSIM) index to evaluate the image quality. The values of RMSE and SSIM for the FBP method are 0.206 and 0.425 respectively, and for the proposed method are 0.063 and 0.985 respectively. Results of reconstruction using the FBP and the proposed method are shown in Fig. 3. These results indicate that the proposed technique enhance the image quality even the small ellipses in the phantom are well reconstructed as shown in the red rectangle in Fig. 3.

B. Synthetic real data

We implemented the proposed technique on synthetic real data for abdominal CT images obtained from medical image database. This database is composed of 40 3D volumes for multiphase abdominal CT images of total size 3.7GB. Each 3D volume is with slice resolution of 512×512 pixels, 1MB size and 0.63mm pixel size. The x-ray power used in scanning the images is 120KeV. We executed the proposed method on 2D slices from the 3D volumes for three patients. The number of views used is 128, which computed with parallel beam projections. The average execution time of the proposed method is 8 minutes. We implemented the proposed method using two steps: the first step is reconstruction with ART algorithm with 10 iteration number. Then, strong smoothing using TV minimization is performed. The reconstruction in the second step is implemented with these parameters: p is 1.1, β is 0.9 for the first patient and 1.4 for the other two patients, Γ is 128 and K is 28, 25 and 31 for the three patients respectively. Furthermore, we reconstructed the same slices using FBP algorithm to be in comparison with the proposed method for the evaluation purpose. Results of reconstruction using the FBP and the proposed method are shown in Fig. 4. RMSE and SSIM were used to evaluate the image quality. The values of RMSE and SSIM for the proposed method and the FBP for the three patients are displayed in Table. 1. It can be observed that the values of RMSE and SSIM indicate that the proposed method is better than the FBP method. These results indicate that the proposed technique produce high image quality. Also, we can observe that the small details in the images are well reconstructed as shown in the red rectangle in Fig. 4.

V. Conclusion

In this work, we introduce using self-prior information in the iterative image reconstruction approach from sparse-view CT. The self-prior information is produced as follow: an initial reconstruction with ART is performed with small numbers of iteration. Then, strong smoothing for ART image using TV minimization is done. After construction the prior information, merging it in the reconstruction process to remove the streak artifacts. The proposed technique is useful compared with methods that use prior information for two reasons. First, there

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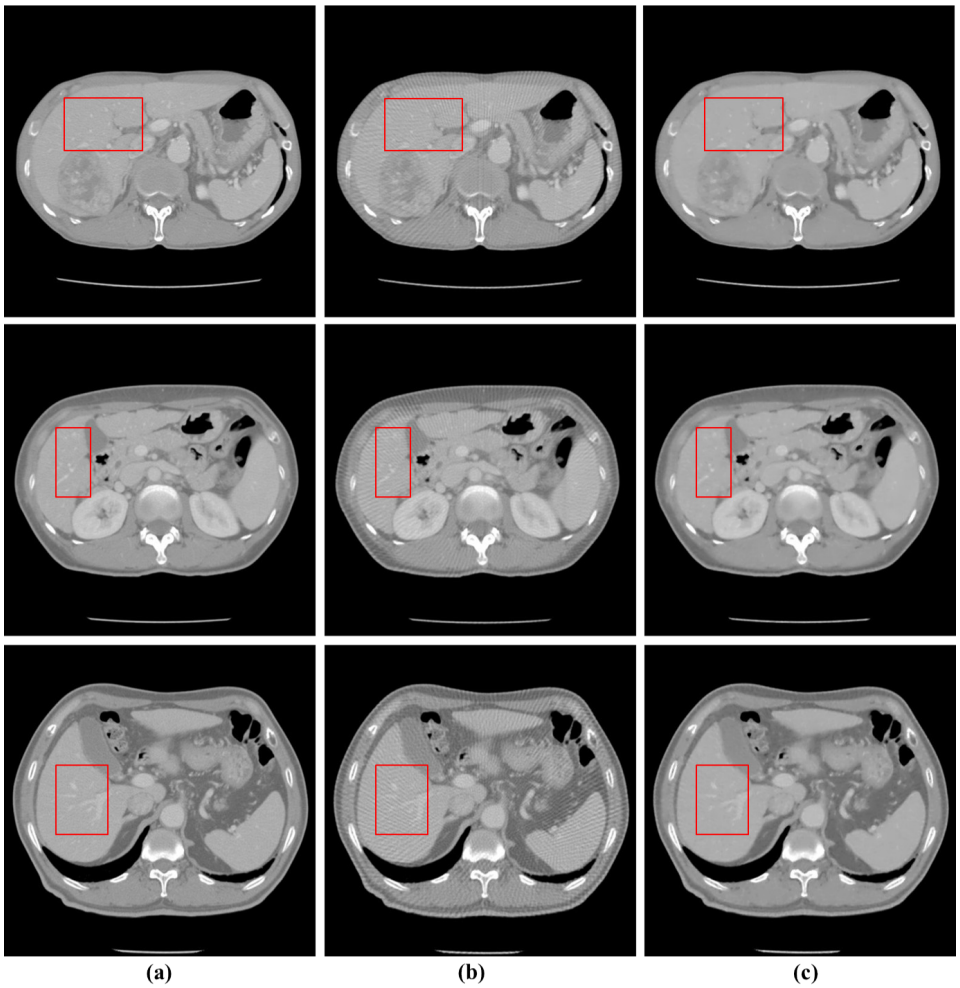

Fig. 4. Central slices of three patients from the abdominal CT images, each row represent different patient. (a) The original slice image, (b) and (c) represent the reconstructed images using FBP and the proposed technique, respectively. The red rectangles in the images indicate that even the small details are well reconstructed. The gray scale is displayed from 0.54 and 1.41 cm^{-1}.

is no longer need to do registration process and consequently this overcomes the limitation of the methods that depend on the accuracy of the registration process. Second, there is no longer need to prepare prior information before the reconstruction. The performed experimental results show that the proposed technique enhances the image quality.

	Patient 1		Patient 2		Patient 3	
	RMSE	SSIM	RMSE	SSIM	RMSE	SSIM
FBP	0.073	0.524	0.067	0.601	0.072	0.523
Proposed	0.020	0.932	0.017	0.945	0.022	0.908

Table 1: RMSE and SSIM values measured using the FBP and the proposed method for the three patients.

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